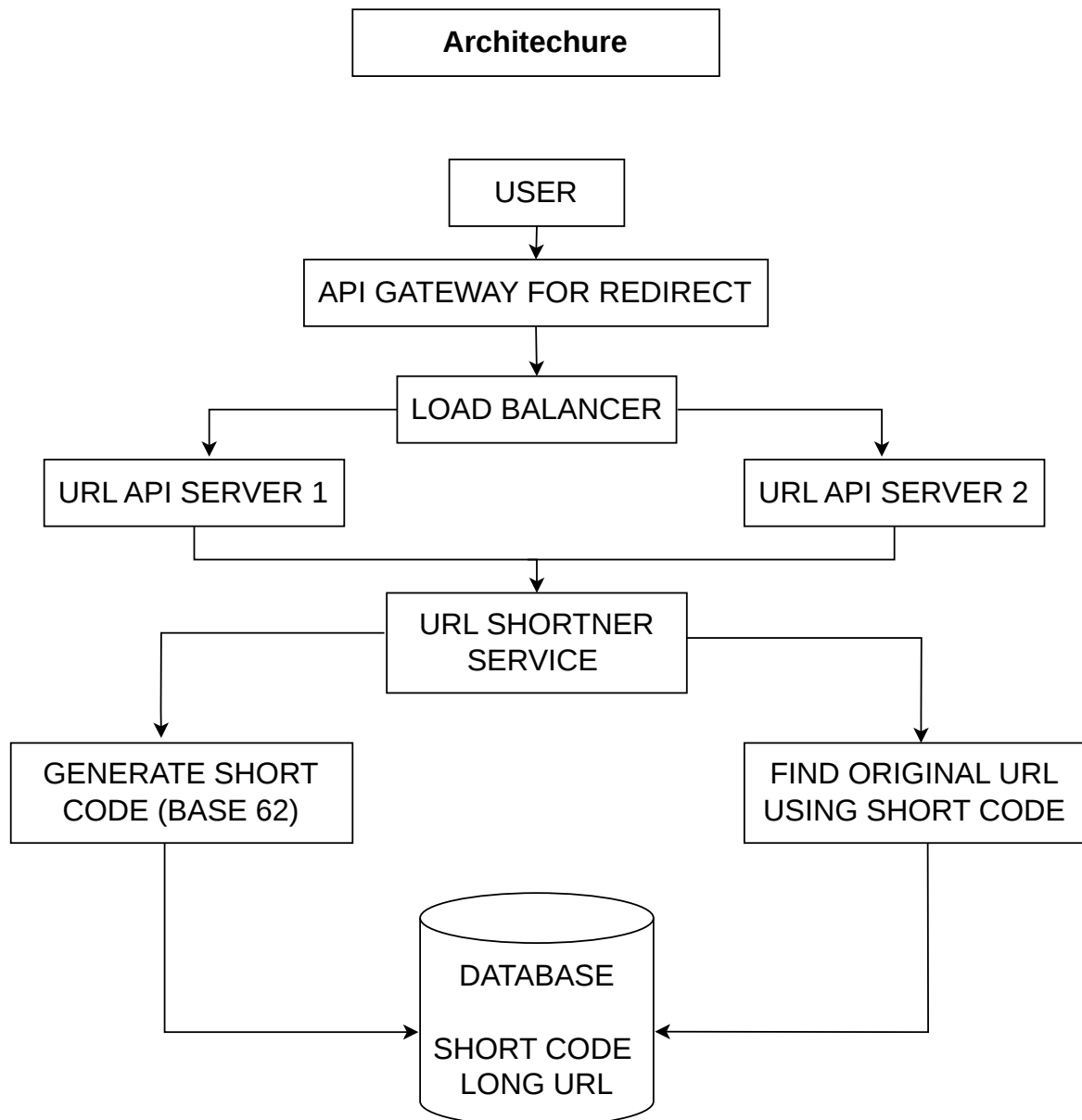
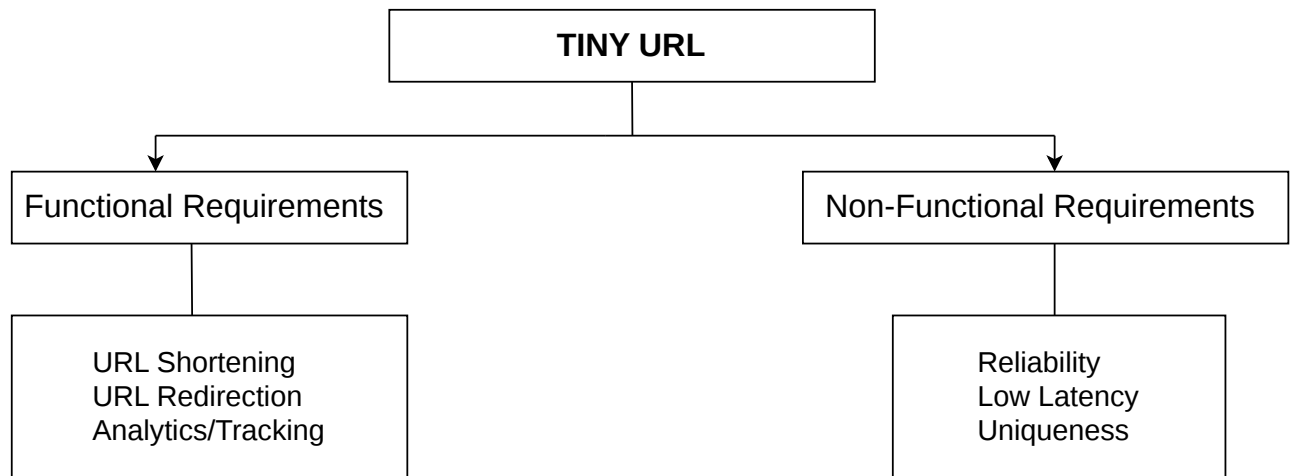




Assumptions

Daily active users : 10 M
1 short URL created per user/day
100 redirects per URL/day

Estimations

- Write Requests/sec = $10M / 86,400 \approx 120$ writes/sec
- Read Requests/sec = $(10M \times 100) / 86,400 \approx 11,600$ reads/sec
- Storage per URL ≈ 500 bytes (long URL + short URL + metadata)
 - Storage/day = $10M \times 500 \text{ B} \approx 5 \text{ GB/day}$
 - Storage/year = $5 \text{ GB} \times 365 \approx 1.8 \text{ TB/year}$
- Short URL Length = 6 Base62 characters ($62^6 \approx 56.8$ billion unique URLs)